

Halogen bonding and other noncovalent interactions involving halogens: a terminology issue†

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Among the various intermolecular interactions involving halogens, halogen bonding refers to those in which halogens behave as acceptors of electron density.

The term *halogen bonding*, or *halogen bond*, has been introduced^{1a} and is used^{1b} for describing any noncovalent interaction involving halogens as acceptors of electron density. The general scheme $D \cdots X \cdots Y$ thus applies to halogen bonding (XB), wherein X is the halogen (Lewis acid, XB-donor), D is any electron-donor (Lewis base, XB-acceptor), and Y is carbon, halogen, nitrogen, *etc.* (Fig. 1). The definition itself sheds light on the nature of XB, which possesses numerous similarities with hydrogen bonding (HB), wherein hydrogen functions as the acceptor of electron density.

In the paper entitled “Multifurcated halogen bonding involving $\text{Ph}-\text{Cl} \cdots \text{H}-\text{CPh}=\text{N}-\text{R}'$ interactions and its relation to idioteloamphiphile layer architecture”, Glaser and co-workers discuss the role of halogen bonding in the crystal structure of 4-chlorobenzaldazine.² They describe the presence of a set of $\text{C}(\text{sp}^2)-\text{H} \cdots \text{Cl}-\text{C}(\text{sp}^2)$ interactions with $\text{H} \cdots \text{Cl}$ distances in the range 3.00–3.41 Å and $\text{H} \cdots \text{Cl}-\text{C}$ angles in the range 79.7–143°, referring to them as an excellent example of multifurcated XB at the *acceptor* Cl. These angles significantly deviate from the typical geometry for halogen bonding (wherein the covalent and noncovalent bonds around the halogen are nearly colinear). They also report that calculations (at the MP2/6–311G** level) predict that all of the

$\text{C}(\text{sp}^2)$ -attached H-atoms are positively charged and a specific relation between the electron density depletion and the attractive nature of the $\text{C}(\text{sp}^2)-\text{H} \cdots \text{Cl}-\text{C}(\text{sp}^2)$ interactions is suggested.

In our opinion, the electronic and geometric features of the $\text{C}(\text{sp}^2)-\text{H} \cdots \text{Cl}-\text{C}(\text{sp}^2)$ contacts shown by the structure of the chlorobenzaldazine are perfectly consistent with a Cl atom which functions as an electron density-donor site. Thus, in the paper by Glaser, the term halogen bonding is used for interactions that are remarkably different from those encompassed by the definition of halogen bonding given in the literature; in fact the electrophile–nucleophile pattern of these $\text{C}(\text{sp}^2)-\text{H} \cdots \text{Cl}-\text{C}(\text{sp}^2)$ contacts is opposite to the one contemplated by this definition. As a confirmation of this terminology misuse, other authors refers to interactions involving $\text{C}(\text{sp}^2)$ -attached H-atoms and halogens (F and I) in the structures of halobenzaldazines as hydrogen bonding.³

It is well-known that the electron density around the halogen nucleus is highly anisotropic so that halogens can serve both as electron-acceptors and -donors (Fig. 2). Theoretical calculations and experimental results consistently show that the electrostatic potential of F remains negative all around the atom, whereas Cl, Br, and I show the emergence of an electropositive crown along the C–halogen axis, which is surrounded by an electroneutral ring and, farther out, an electronegative belt.⁴ The size of this electropositive crown increases with the polarizability of the halogen, for this reason iodocarbons give rise to the strongest $\text{D} \cdots \text{I}-\text{C}$ XB, whereas F is more likely to serve as HB-acceptor. In keeping with the electrophile/nucleophile model for the interactions involving halogen atoms, the contacts towards the pole (electrophilic end) are those wherein the halogen acts as an electron-deficient site,⁵ while contacts towards the equatorial (nucleophilic end)

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† Comment on “Multifurcated halogen bonding involving $\text{Ph}-\text{Cl} \cdots \text{H}-\text{CPh}=\text{N}-\text{R}'$ interactions and its relation to idioteloamphiphile layer architecture” by R. Glaser, R. F. Murphy, Y. Sui, C. L. Barnes and S. H. Kim, *CrystEngComm*, 2006, 8, 372

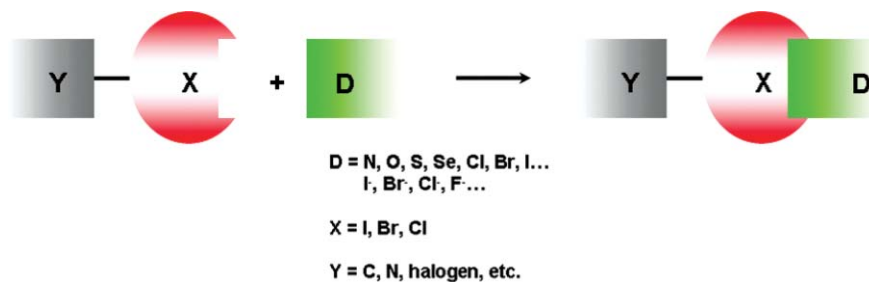


Fig. 1 General scheme of the donor/acceptor sites typically involved in XB.

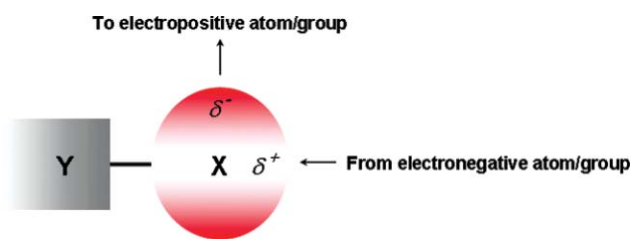


Fig. 2 Pattern of the electronic distribution around halogen atoms (Y as in Fig. 1) (adapted with changes from ref. 5b).

are those in which the same halogen acts as an electron-rich site. The term halogen bonding addresses exclusively the former contacts and its usefulness relies solely on the identification of a specific subset of the numerous and diverse noncovalent interactions halogens can give rise to.

While we acknowledge that the terminology used to name noncovalent interactions given by halogen atoms has to be as unifying as possible, it is our opinion it has always to be done in keeping with the electrophile/nucleophile role the halogen atom plays. The use of the term halogen bonding has to be

limited to those interactions wherein halogens function as acceptors of electron density. Its use also to address interactions in which the halogen works as an electron-donor contrasts with both the definition and the tendency, well-documented in the literature, to name $D\cdots H-Y$ interactions as hydrogen bonds.^{3,5} Glaser's use of the term halogen bonding is thus conceptually misleading.

References

- (a) J. M. Dumas, M. Gomel and M. Guerin, in *The Chemistry of Functional Groups, Supplement D*, ed. S. Patai and Z. Rappoport, John Wiley & Sons Ltd, New York, 1983, chap. 21, p. 985; (b) P. Metrangolo, H. Neukirch, T. Pilati and G. Resnati, *Acc. Chem. Res.*, 2005, **38**, 386.
- R. Glaser, R. F. Murphy, Y. Sui, C. L. Barnes and S. H. Kim, *CrystEngComm*, 2006, **8**, 372.
- V. R. Vangala, A. Nangia and V. Lynch, *Chem. Commun.*, 2002, 1304–1305; C. Glidewell, J. N. Low, J. M. S. Skakle and J. L. Wardell, *Acta Crystallogr., Sect. C*, 2005, **61**, o312.
- P. Auffinger, F. A. Hays, E. Westhof and P. Shing Ho, *Proc. Natl. Acad. Sci. U. S. A.*, 2004, **101**, 16789.
- (a) L. Brammer, E. A. Bruton and P. Sherwood, *Cryst. Growth Des.*, 2001, **1**, 277; (b) A. Dey, R. K. R. Jetti, R. Boese and G. R. Desiraju, *CrystEngComm*, 2003, **5**, 248.